india-tax-engine

Final product requirements and architecture document · Version 1.0 · 9 November 2026

Purpose. This is the last planning document. It merges the first product brief with the product review and roadmap, resolves the points where they disagree, refreshes the research, and fixes the scope, architecture, API and phases for a 13-hour build. After this we move straight to building.

1. Decision summary

Question	Decision
What are we building?	A deterministic, offline, dependency-free TypeScript library that calculates Indian income tax for one narrow, clearly documented taxpayer scenario and returns a traceable, explainable result.
Verdict	**BUILD**, as a pre-1.0 prototype (v0.x) with a narrow promise. Strong learning value and a real reuse case in PolicyLens AI. Demand and downloads remain unvalidated.
Time box	13 hours. This covers Phases 0 to 3 below. Phases 4 to 6 happen after the college submission.
Supported in v1	Resident individual, below 60, salary income only; new and old regime; tax periods 2025-26 and 2026-27; standard deduction, rebate, marginal relief, 4% cess, statutory rounding; regime comparison.
Explicitly not in v1	Surcharge (taxable income above ₹50,00,000 returns *unsupported*), capital gains, HRA, house property, business income, NRIs, senior citizens, GST, other countries, AI or scraping inside the package.
Main differentiators	Source-linked rules, versioned rule sets reproducible per result, explicit support matrix, structured calculation trace, exact money handling.

2. Problem, promise and positioning

Problem. Developers who need Indian income-tax logic today copy blog formulas, embed a website calculator, or write the rules from scratch. Rules changed recently: the Income-tax Act, 2025 replaces the 1961 Act from 1 April 2026, and “financial year / assessment year” becomes “tax year”. Hardcoded logic breaks silently, and simple calculators often miss the section 87A marginal-relief rule just above ₹12,00,000.

Product promise. For a supported taxpayer scenario and a specified tax period, calculate tax using versioned, source-linked rules and return an explainable result that developers can test and audit. Anything outside the supported scenario is rejected explicitly, never silently ignored.

Positioning statement to test. A developer-first, auditable Indian income-tax calculation engine with versioned statutory rules, deterministic outputs and transparent calculation traces.

What we do not claim. We do not claim to be first, only, most accurate, or automatically up to date. JSON rule files make changes easier to implement; they do not discover or verify law. Updates are a reviewed, human workflow. The output is an informational estimate, not tax advice and not suitable for filing.

3. Target users

User	Need	Fit
Student and indie developers building salary or finance apps	Correct tax logic without learning the law first	One typed function call with a readable trace
Small HR, payroll and fintech prototypes	A transparent estimate employees can follow	Deterministic output, support matrix, versioned rules
AI apps and agents	A calculation tool instead of a model guessing arithmetic	Pure function; a thin adapter can be added later
PolicyLens AI (our own project)	Tax-impact and eligibility component	Direct reuse, but a demo, not independent adoption

Not for: chartered accountants, official return filing, or legally binding advice.

4. Evidence refresh (9 November 2026)

4.1 Competition

npm registry searches and package metadata checked on 9 November 2026. Registry search is fuzzy, so this is a lead list, not a census. Re-check before publishing and before any public comparison.

Package / category	Observation	Implication
itr-agent (npm)	Local-first MCP server for Indian income tax: compute, compare regimes, plan advance tax. One version published, 21 July 2026, MIT.	Overlaps on calculation and regime comparison, but targets AI agents. We stay a plain library.
calcsuite-react (npm)	Decimal-exact, offline calculator library for React, India and US, about 55 calculators. 11 versions, MIT.	Breadth and UI. We compete on depth, provenance and traceability.
india-finance-calc	Named in the review. No matching package found on the npm registry (zero versions).	May be GitHub-only or unpublished. Unverified. Do not cite as an npm competitor.
Other-country engines	hmrc-income-tax (UK), dutch-tax-income-calculator (year-based versions), nigeria-paye, bd-tax; also @trajectonomics/tax, a multi-jurisdiction engine.	Year-versioned national tax engines have an audience. A multi-jurisdiction engine already exists, so global expansion is not a differentiator.
Official calculator and filing portals	Authoritative for end users.	We serve developers who need an embeddable library.
GST utilities	Many packages (gstin-utils and others).	Out of scope.

Name check. india-tax-engine, india-tax-engine-core and in-tax-engine returned 404 (free) on 9 November 2026. Re-check on publish day.

4.2 Market and demand

The law change in April 2026 and the new-regime default make the problem current. Demand for a niche developer library is unproven, and npm download statistics were not accessible during research, so no demand figure is claimed. Success is judged on correctness, documentation and real integration, with downloads as a bonus.

5. Where the two earlier documents disagree, and how we resolve it

Earlier statement	Issue	Resolution in this document
“Rules are JSON, so a Budget change is just a new file.”	JSON stores rules; it cannot discover or verify law.	Reworded. Rule files make changes easier to implement. A manual review process (Phase 4 automates detection later) owns correctness.
“Add surcharge if time remains.”	Implying support for an unfinished rule is unsafe.	Hard ceiling: total income above ₹50,00,000 returns status *unsupported*. No partial surcharge.
Single {income} field.	Will not extend to more income heads.	Nested input with income.salary only in v1. Other heads are rejected, not ignored.
“Integer paise” money.	Statute also requires rounding to the nearest ₹10 at two stages.	Both: internal integer paise for arithmetic, plus statutory ₹10 rounding of total income and of payable tax.
12 to 13 hour estimate.	Fine for a prototype, not for production tax software.	Ship as v0.x and label it a prototype. Production readiness is Phases 4 and 5.
“No library combines these features.”	Cannot be proven; competitors keep appearing.	Dropped. Positioning stays factual and dated.
Downloads as proof of success.	Downloads include bots and CI.	Downloads are one metric among several (section 14).
Large repository layout with scripts and docs folders.	Too much for 13 hours.	Trimmed layout (section 8.3). Heavier folders are deferred.
Blogs as rule sources.	Blogs conflict with each other.	Every rule carries a verification label. Release needs primary-source confirmation (section 7).

6. Feature disposition

Feature	Decision	Phase	Note
Deterministic pure calculation core	Keep	1	No network, no clock, no randomness
Integer-paise arithmetic and ₹10 statutory rounding	Keep	1	Two rounding stages; tested separately
Explanation trace (machine-readable) plus text helper	Keep	1	Steps are data; prose is a separate function
Regime comparison	Keep	1	Reports eligibility; propagates unsupported
Result status complete / unsupported	Keep	1	Reserved status *incomplete* arrives with more income heads
Rule metadata and provenance	Keep (minimal)	2	Source, provision, retrieved date, verification label, ruleset version
Golden, boundary, interaction and regression tests	Keep	1 and 2	Release gate in section 10
Rule schema validation script	Keep (minimal)	2	Simple validator; no heavy tooling
ESM and CJS dual build, typings, npm provenance	Keep	3	Verify with a real consumer project
Playground, framework examples	Defer (one example only)	3 / later	One Node and one Next.js snippet in the README
Surcharge	Defer	5	Marginal relief at each surcharge threshold is real work
HRA, house property, capital gains, business income, NRI, seniors	Defer	5	Each needs its own legal analysis and tests
Update monitoring, AI-drafted change proposals	Defer	4	Separate tool; human approval; never at runtime
MCP or AI adapter	Defer	5	itr-agent already covers MCP; adapter only if PolicyLens needs it
Multi-jurisdiction support	Defer	6	Shared infrastructure only; rules stay per country
Hosted API, paid monitoring	Not planned	-	Needs privacy and security design first

7. Rule baseline and verification status

Honest status. The figures below are corroborated by multiple secondary sources (tax-advisory sites and explainers, 9 November 2026 research). The rounding rule was checked against an Income Tax Department FAQ. I could not open the primary Act or Finance Act text for the slab tables. **Phase 0 must confirm every row against primary sources before any rule is marked verified.**

Rule	New regime (2025-26 and 2026-27)	Old regime, below 60	Status
Slabs	0 to 4L nil; 4 to 8L 5%; 8 to 12L 10%; 12 to 16L 15%; 16 to 20L 20%; 20 to 24L 25%; above 24L 30%	0 to 2.5L nil; 2.5 to 5L 5%; 5 to 10L 20%; above 10L 30%	Secondary-corroborated
Standard deduction (salary)	₹75,000	₹50,000	Secondary-corroborated
Rebate (87A / s.156)	Total income up to ₹12,00,000; up to ₹60,000; resident only	Total income up to ₹5,00,000; up to ₹12,500	Secondary-corroborated
Marginal relief on rebate	Yes: tax above ₹12,00,000 limited to income above the limit	Sources conflict (most say none; one says it may apply)	**Open conflict**
Health and education cess	4% on tax after rebate and relief	4%	Widely stated

Rule	New regime (2025-26 and 2026-27)	Old regime, below 60	Status
Rounding	Total income and payable tax rounded to nearest ₹10 after ignoring paise (ss.288A and 288B; s.516 of the 2025 Act)	Same	Primary FAQ for 1961 Act; confirm s.516
Surcharge	Above ₹50L (10%, 15%, 25% capped)	Above ₹50L (to 37% above ₹5 crore)	Not implemented; ceiling ₹50L
Legal mapping 2026-27	Rebate moves from s.87A (1961 Act) to s.156 (2025 Act); new-regime provision reported as s.202	Same Act	Confirm section numbers

Notes. Sources report the slab numbers and the ₹12 lakh rebate are unchanged for 2026-27 compared with 2025-26, with only the legal framework and terminology changing. So the two rule files will carry the same numbers but different legal references, and a regression test asserts identical results. Under the new regime, sources say only the standard deduction and employer NPS contributions are permitted deductions; v1 treats NPS as unsupported and rejects it. The old-regime marginal-relief conflict must be settled from primary text before the old-regime rule is marked verified; until then the engine follows the majority reading (no relief) and flags it in the rule status.

8. Architecture

8.1 Layers

Layer	Responsibility	Key rules
Public API	calculateTax, compareRegimes, explain	Stable typed contracts; documented defaults
Input validation	Types, ranges, known period and regime	Malformed input throws TaxInputError with a code
Support check	Scenario matrix	Unsupported scenarios return status unsupported with reasons; never silently ignored
Rule registry	Select ruleset by period and regime	Bundled, immutable, offline; carries rulesetVersion
Calculation core	Ordered, pure stages	No network, clock or randomness
Money	Integer paise, ₹10 rounding	Safe-range limit; rounding tested per stage
Explanation	Structured steps plus text renderer	Data and prose kept separate

8.2 Calculation pipeline and product flow

```
Caller: calculateTax(input)
  1. Validate types and ranges            -> TaxInputError on malformed input
  2. Support check                        -> status "unsupported" if outside matrix
     (resident, individual, below 60, salary only, no unsupported deductions)
  3. Load ruleset (period + regime)       -> rulesetVersion, appliedRules
  4. Convert rupees to paise
  5. Gross salary - standard deduction
     - declared Chapter VI-A (old regime only; assumed valid, warned)
     = total income (floor at zero)
  6. Round total income to nearest 10 rupees (ignore paise first)
  7. Support ceiling: total income > 50,00,000 -> "unsupported" (no surcharge)
  8. Slab tax, layer by layer (rates stored in basis points)
  9. Rebate (income <= limit): min(slab tax, max rebate)
 10. Marginal relief (new regime, income above limit):
     tax = min(slab tax, income - limit)
 11. Cess = 4% of tax after rebate / relief
 12. Round payable tax to nearest 10 rupees
 13. Return { status:"complete", totalIncome, taxPayable, steps[],
              appliedRules[], rulesetVersion, assumptions[], warnings[] }

compareRegimes(input): run the flow for "new" and "old" with the same facts,
return both results, the cheaper regime and the difference.
```

8.3 Repository layout (trimmed for 13 hours)

```
india-tax-engine/
  src/
    api/          calculateTax, compareRegimes, explain, public types
    validation/   input checks, support matrix
    core/         calculation stages (pure functions)
    money/        paise helpers, rounding
    rules/        rule types, registry, loader
  rules/india/
    2025-26.json  2026-27.json     (metadata + provenance inside)
  tests/          golden/  boundary/  interaction/  regression/  negative/
  docs/           supported-scenarios.md  sources-and-methodology.md
  scripts/        validate-rules.ts
  examples/       node.js  nextjs.md
  README.md  LICENSE  CHANGELOG.md  package.json  tsconfig.json
Deferred: generate-change-report/, release-process.md, playground/
```

8.4 Rule file shape (illustrative, values pending Phase 0 verification)

```
{
  "rulesetVersion": "IN-2026-27-new-r1",
  "jurisdiction": "IN", "taxPeriod": "2026-27", "framework": "IT_ACT_2025",
  "regime": "new",
  "effective": { "from": "2026-04-01", "to": "2027-03-31" },
  "status": "draft",                      // draft | verified | retired
  "slabs": [ { "upTo": 400000, "rateBps": 0 }, { "upTo": 800000, "rateBps": 500 },
             /* ... */ { "upTo": null, "rateBps": 3000 } ],
  "standardDeduction": 75000,
  "rebate": { "incomeLimit": 1200000, "max": 60000, "marginalRelief": true },
  "cessBps": 400,
  "rounding": { "totalIncome": 10, "payable": 10 },
  "support": { "maxTotalIncome": 5000000 },
  "sources": [ { "ruleId": "IN-26-N-SLAB", "provision": "to confirm",
                 "url": "https://www.incometax.gov.in/", "retrievedOn": "YYYY-MM-DD",
                 "verification": "secondary-corroborated" } ]
}
```

9. Public API

```
type Regime = "new" | "old";

interface TaxInput {
  jurisdiction?: "IN";                          // default "IN"
  taxPeriod: "2025-26" | "2026-27";
  regime: Regime;
  taxpayer: { category: "individual"; residentialStatus: "resident";
              ageBand: "below-60" };
  income:   { salary: number };                 // INR, gross annual, max 2 decimals
  deductions?: { chapterVIA?: number };         // old regime only; declared total
}

type TaxResult = CompleteResult | UnsupportedResult;

interface CompleteResult {
  status: "complete"; amountUnit: "INR"; taxPeriod: string; regime: Regime;
  rulesetVersion: string; totalIncome: number; taxPayable: number;
  steps: { order: number; code: string; ruleId: string; amount: number }[];
  appliedRules: { ruleId: string; ruleVersion: string }[];
  assumptions: string[]; warnings: { code: string; message: string }[];
}
interface UnsupportedResult {
  status: "unsupported";
  reasons: { code: string; message: string; field?: string }[];
}
```

- **Units:** all inputs and outputs are INR rupees; arithmetic is integer paise internally. The result carries amountUnit to make this unmistakable. Inputs above a documented safe limit are rejected.

- **Errors:** malformed input throws TaxInputError (stable code). Valid input outside the support matrix returns status unsupported. No plausible total is ever returned when a material input is unsupported.
- **Reproducibility:** every result carries rulesetVersion so a past result can be reproduced.
- **Privacy:** the API asks only for facts needed for calculation. No personal identifiers.
- **Chapter VI-A deductions:** a single declared total, applied in the old regime only, never validated against section caps. The result adds an assumption and a warning saying so.

10. Testing and release gate

10.1 Golden cases (hand-computed from the baseline in section 7)

These must be re-checked against the official Income Tax Department calculator and primary sources in Phase 0. They are my arithmetic, not an independent certification.

Case	Regime	Total income	Slab tax	After rebate / relief	Cess	Payable
Salary 12,75,000	new	12,00,000	60,000	0 (rebate)	0	0
Salary 12,00,000	new	11,25,000	52,500	0 (rebate)	0	0
Salary 12,85,000	new	12,10,000	61,500	10,000 (relief)	400	10,400
Salary 13,00,000	new	12,25,000	63,750	25,000 (relief)	1,000	26,000
Salary 15,00,000	new	14,25,000	93,750	93,750	3,750	97,500
Salary 5,50,000, no deductions	old	5,00,000	12,500	0 (rebate)	0	0
Salary 6,00,000, no deductions	old	5,50,000	22,500	22,500	900	23,400

10.2 Test categories

- **Boundary:** one rupee below, at, and above ₹4L, 8L, 12L and the ₹12L rebate limit (new), and ₹5L (old).
- **Interaction:** rebate with marginal relief, deduction with regime, rounding with rebate threshold.
- **Rounding:** paise ignored first, then nearest ₹10, for total income and for payable tax separately.
- **Regression:** 2025-26 and 2026-27 give identical results for identical inputs; adding a new period never changes an old one.
- **Property:** non-negative amounts, steps sum to totals, same input gives same output.
- **Negative:** unknown period or regime, NaN, negative, excessive magnitude, unsupported fields, income above ₹50L.

Release gate. Do not publish until: every rule has a source and verification label; every advertised scenario has tests; boundary and interaction tests pass; regression tests are stable; expected values were cross-checked against an independent calculator. A green suite is necessary but not sufficient, because the expected values themselves could be wrong.

11. Phases and the 13-hour plan

Seven phases exist in total. **Four phases (0 to 3) fit inside the 13 hours and are what we submit.** Phases 4 to 6 are future work after the assignment.

Phase	Scope	Hours	Deliverable	Exit gate
0 Validate	Confirm every baseline rule against primary sources; settle old-regime marginal-relief conflict; re-check npm name and competitors; verify golden cases against the official calculator; set up repo, TypeScript, test runner	1.0	Source list; confirmed rule values; repo skeleton	Every rule row has a source; conflicts resolved or flagged
1 MVP	Money and rounding (1.0); types, validation and support matrix (1.0); rule registry and 2 rule files (1.5); slab, rebate, relief, cess pipeline (2.0); explanation trace and regime comparison (1.5)	7.0	Working calculateTax and compareRegimes	All golden cases pass; unsupported cases rejected
2 Trust	Provenance metadata in rule files, rule validator script, boundary, interaction, regression and negative tests, assumptions and warnings	2.5	Verified-labelled rules; full test suite	Release gate in section 10 satisfied
3 Adoption	README with 30-second quick start, support matrix, limitations and disclaimer; ESM and CJS build, typings; consumer install test; npm pack check; publish v0.1.0	2.5	Published package; README; sources doc	Fresh project installs and runs the quick start
Total		**13.0**		

Time-box rule. If a phase overruns, cut from the end of that phase, never from tests or source verification. Cut order: examples, extra README sections, then optional old-regime deduction handling (reject it as unsupported instead).

Post-submission phases

Phase	Scope	Exit gate
4 Update workflow	Official-source monitoring, proposed diffs, human review, test generation, release automation	A tax update is processed safely and reproducibly
5 Expansion	Surcharge, then additional income heads and taxpayer categories, each with its own legal analysis and tests; optional AI or MCP adapter	Each addition has a support contract and verified cases
6 International	One new jurisdiction as a separate rule and calculation module	Second jurisdiction added without weakening India

12. Assignment submission pack

- Published npm package link and version (v0.1.0), with the live package page
- GitHub repository with a clean commit history per phase
- Passing test output screenshot, including boundary cases (₹12,00,000 and ₹12,10,000)
- README, supported-scenarios and sources-and-methodology documents
- Fresh-install proof: a new folder, npm install, quick-start output
- This PRD, plus a short reflection on the decisions in section 5
- Download statistics as supporting evidence only, never inflated

Viva preparation. Be ready to explain: why rules are data; why integer paise; the two rounding stages; marginal relief with the ₹12,10,000 example; why surcharge is out of scope; why the package never calls the network; why unsupported input is rejected instead of ignored.

13. Risk register

Risk	Severity	Mitigation
Incorrect tax calculation	Critical	Primary-source verification, golden and boundary tests, independent cross-check, v0.x labelling
Unsupported income silently ignored	Critical	Strict schema, support matrix, unsupported status
Missed law change or effective date	Critical	Effective dates in every rule; manual review checklist now, automated monitoring in Phase 4
Conflicting secondary sources	High	Verification labels; primary text decides; unresolved conflicts flagged in rule status
Rounding and threshold interactions	High	Separate rounding tests; boundary tests at each threshold
Low adoption	High	Honest metrics; PolicyLens integration labelled as a demo; one clear example
Legal and trust expectations	High	Disclaimer; no claim of official endorsement or filing suitability
Scope creep past 13 hours	Medium	Time-box rule in section 11
Packaging and runtime incompatibility	Medium	Test ESM and CJS in a real consumer project; Node versions listed
Supply-chain exposure	Medium	Zero runtime dependencies; pinned dev tooling; inspect pack contents

14. Success metrics

Metric	Use
Calculation agreement with verified cases	Core quality metric
Install and build success in a fresh project	Package works for others
Declared dependents and real integrations	Stronger than stars or downloads
Issues, fixes and external contributions	Community engagement and maintenance
Rule update lead time	Official publication to verified release (from Phase 4)
npm downloads, GitHub stars	Visibility only; may include bots and CI

15. Decision log and open items

Decisions made

- Name stays india-tax-engine for v1; revisit before any second jurisdiction (the name is free as of 9 November 2026).
- Rules are bundled with the package version, offline and immutable. A separate rules package is a later option.
- Public money unit is INR rupees; internal arithmetic is integer paise.
- Tax period identifiers are 2025-26 and 2026-27, mapped to IT_ACT_1961 and IT_ACT_2025 frameworks.
- Project stays open source and community-run; no hosted API or paid monitoring is planned.
- No AI, scraper or database inside the package. Any update tooling stays separate.

Open items to close in Phase 0

- Confirm all slab, deduction, rebate and cess values against primary sources for both periods and both regimes.
- Resolve whether marginal relief applies to the old-regime rebate; record the primary citation.

- Confirm the 2025 Act section numbers (rebate s.156, new-regime provision, rounding s.516).
- Confirm employer NPS and other permitted deductions under the new regime, then keep them unsupported or add them.
- Cross-check all golden cases against the official calculator.
- Re-run the npm name and competitor check on the day of publication.

16. Sources and research notes

- npm registry searches and package metadata, 9 November 2026 (competition, name availability).
- Secondary tax explainers on slabs, standard deduction, rebate and marginal relief for 2025-26 and 2026-27 (for example Taxaj, CA Club India, TaxGuru, CalcGuru, Maurya Tech, Canara HSBC Life); used as corroborating leads only.
- Surcharge rates: TaxGuru, Zoho Payroll, Taxaj and others (secondary).
- Rounding: Income Tax Department FAQ on section 288A, TaxTMI and TaxGuru notes on section 288B and Clause 516 / Section 516 of the Income-tax Act, 2025.
- Income-tax Act, 2025 transition and section mapping: Taxscan, India Briefing, Taxaj, TaxGarden (secondary).
- The supplied product brief and the product review and roadmap PDF.
- Primary sources to rely on for final values: incometax.gov.in, incometaxindia.gov.in, the Act, Finance Acts, rules and notifications.

Legal boundary. This is a product and engineering document, not tax, legal or financial advice. Tax figures here are research baselines pending primary-source verification and must not be presented as verified until Phase 0 is complete.